Structural certificates for the Sp(4, 7) atlas table

Private L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and named structural registry, all 603 rows of the current companion table headed* $\mathrm{Sp}(4,7)$ *have the displayed structural descriptions. Here the ambient group is* $\mathrm{PSp}_4(7)$*, of order* 138297600*. An extension description asserts the indicated actual normal subgroup, quotient and splitting behaviour; it does not classify an abstract group uniquely by its label.*

Exact source and embeddings. The Connor–Leemans table is dated 7 June 2013. All 28 native pages were read. Its numerical catalogue, source-row bijection and weighted maximal inclusions are acquired from the separately reviewed projective seed and completed descent. That input retains the explicit classical-maximal-cover premise; the present application supplies no replacement classification. The ordered 400-point ambient generators are identical throughout the initial projective seed, completed state and retained subgroup file. Every proper subgroup is given by literal embedded permutation generators. No printed constructor name substitutes for a proper embedding.

The native source fixes powers and punctuation before any group proposal. In particular rows 25 and 410 contain cyclic 42, rows 40,460,473 cyclic 24, and row 458 cyclic 25; elsewhere the flattened strings may denote 4^2 or 2^4. The atoms 7^3 and 7^{1+2} are distinct. Parentheses around the single atoms Q_{16}, D_{16}, D_{14} are redundant and their original text is retained. Every parsed tree is joined to the exact native assertion, row, order and acquired embedded generators.

Reusable finite certificate. The registry contains 332 acquired models and 242 new models: 174 split, 41 nonsplit, 25 direct, one elementary abelian and one extraspecial model. The additional $L_2(49)$ atom is the exact named top from the accepted small-order-domain application, not an inference from its order. All factor references are acyclic. Complete literal generator closures and identity-preserving maps respecting every right-generator edge prove all model and occurrence isomorphisms by induction on words.

For a displayed split extension the reader checks $N \triangleleft H$, $K \leq H$, $N \cap K = 1$ and $|N||K| = |H|$. A colon also has a nontrivial actual action; direct factors commute. Elementary abelian factors have a commuting prime-order generating basis and the exact order. The new extraspecial factor has order 7^3, generators x, y, and $z = [x, y]$ of order seven with $Z(H) = \langle z \rangle$ and $x^7, y^7 \in \langle z \rangle$. Hence $H/Z(H) \cong C_7^2$ and $H' = Z(H)$; maximal-subgroup quotients give $\Phi(H) = Z(H)$, proving the unsigned extraspecial assertion used by the source.

The two central-product certificates. The nonsplit groups underlying rows 8 and 11 have a central kernel $N \cong C_2$ and quotient $L_2(7) \times L_2(7)$. Their certificates exhibit actual subgroups A, B of order 336 with

$$[A, B] = 1, \qquad A \cap B = N, \qquad |A||B| = |H||N|.$$

Thus $H = AB$. Complete right N-cosets in each factor give two regular quotient actions, each independently transported to the acquired $L_2(7)$ model. The map $(A/N) \times (B/N) \to H/N$ induced by multiplication is well defined and surjective. Its kernel is trivial by $A \cap B = N$, so it is the asserted direct-product isomorphism.

Actual witnesses $x, y \in A$ have $1 \neq [x, y] \in N$, and $N \leq Z(H)$. Their images commute in H/N, while every pair of lifts has the same nonidentity commutator. A complement would supply commuting lifts, which is impossible. This last argument is the previously accepted commuting-lifts criterion, originally suggested by the root collaborator; its authorship is retained. The new work here is the explicit central product and its factor-quotient identifications. The reader also checks all four shifted pairs of lifts literally.

The other dots use the acquired complete prime, cyclic-six, S_3, abelian-quotient or general Cayley-lift criteria. For the latter, every choice of lifts of the identified quotient's ordered generators is followed along its full Cayley graph. A consistent assignment is a section of the quotient map; conversely a complement supplies such an assignment. One inconsistent edge for every lift choice proves nonsplitting. For S_3, first retain only lifts with $a^2 = 1$ and $b^3 = 1$, then test $(ab)^2$ on their Cartesian product. Every discarded pair already fails a necessary relation; the result and total original-pair count are unchanged. Exact old/new small truth tables test this factoring. Failed complement or isomorphism searches are merely proposal outcomes, never certificates of nonexistence.

Exact larger-degree encoding. For $257 \leq d \leq 512$, each image a_i is stored in two byte planes $a_i = \ell_i + 256h_i$, where $h_i \in \{0, 1\}$. For right-action composition $(ab)_i = b_{a_i}$, each plane of b is split into its first 256 and remaining entries. Two byte lookups indexed by ℓ_i, selected by the all-zero or all-one byte mask determined by h_i, give exactly both planes of b_{a_i}. The selection is bitwise and introduces no carries. Canonical encoding preserves equality and hashing; iteration, inversion and JSON serialization recover the original point images. Other degree ranges retain their acquired representation. Independent tuple multiplication, sparse high-label permutations and actual source-generator pairs check this implementation boundary, including an explicit regular Heisenberg group over $\mathbb{F}_7$.

Validation and scope. Bounded proposal attempts and their failures are preserved individually. An early driver returned before its intended calls; other attempts reached their caps. Later passes used exact earlier positive witnesses and changed only the remaining witness proposal strategy. All 602 proper-row certificates now exist, with the top acquired by its named construction. The final literal reader and damage controls are replayed in ordinary and optimized Python. All source joins, embeddings, quotient maps and lift choices are checked independently of proposal success messages.

No new ambient subgroup census, historical-byte identity, unique interpretation of extension labels, whole-source admission, question resolution, formalization or measurement is claimed. The remaining larger source-associated tables stay in scope.